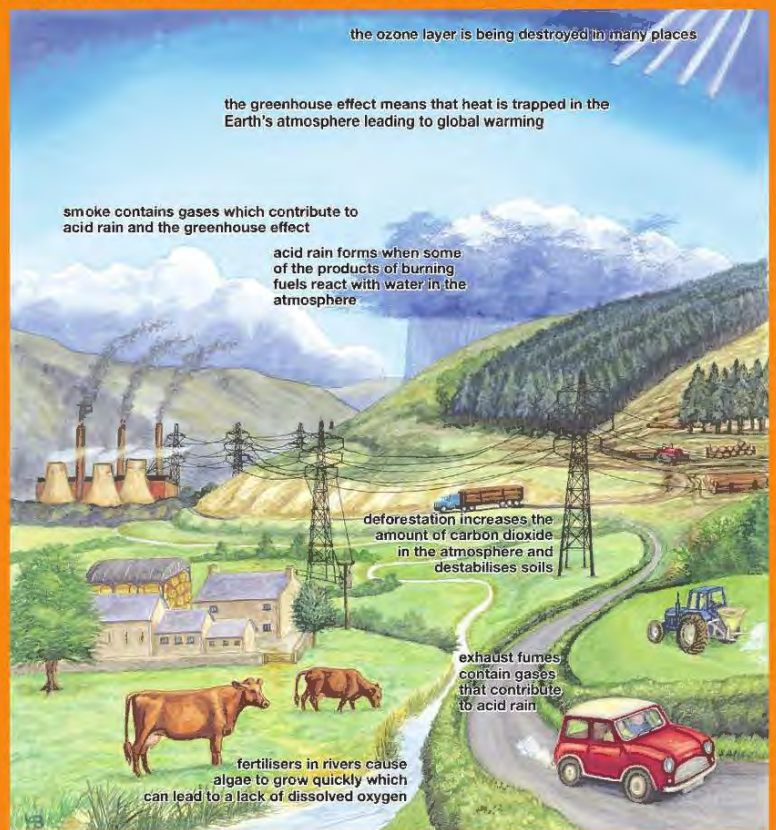
15 HUMAN INFLUENCES ON THE ENVIRONMENT

Humans have intelligence far beyond that of any other animal on Earth. This chapter looks at the ways in which we have used our intelligence to influence natural environments, and some of the problems we have caused.



▲ Figure 15.1 How the actions of humans influence the environment.

LEARNING OBJECTIVES

- Describe how glasshouses and polythene tunnels can be used to increase the yield of certain crops
- Understand the effects on crop yield of increased carbon dioxide and increased temperature in glasshouses
- Understand how the use of fertiliser can increase crop yield
- Understand the reasons for pest control and the advantages and disadvantages of using pesticides and biological control with crop plants

BIOLOGY ONLY

- Understand the methods used to farm large numbers of fish to provide a source of protein

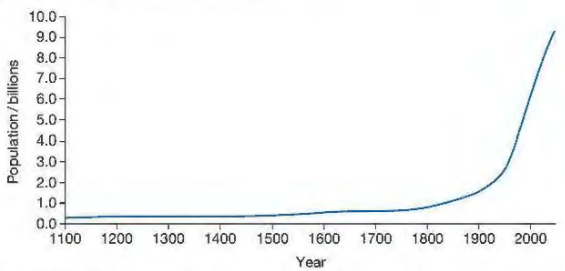
- Understand that water vapour, carbon dioxide, nitrous oxide, methane and CFCs are greenhouse gases
- Understand how human activities contribute to greenhouse gases
- Understand how an increase in greenhouse gases results in an enhanced greenhouse effect and that this may lead to global warming and its consequences
- Understand the biological consequences of pollution of air by sulfur dioxide and carbon monoxide

BIOLOGY ONLY

- Understand the effects of deforestation, including leaching and soil erosion; and disturbance to evapotranspiration, the carbon cycle, and the balance of atmospheric gases

- Understand the biological consequences of pollution of water by sewage
- Understand the biological consequences of eutrophication caused by leached minerals from fertiliser

Since humans first appeared on Earth, our numbers have grown dramatically (Figure 15.2). The secret of our success has been our intelligence. Unlike other species, we have not adapted to one specific environment, we have changed many environments to suit our needs.



▲ Figure 15.2 The rate of increase of the human population has been particularly steep over the last 200 years. The curve has been extended to 2050, showing that the population is expected to reach 10 billion by that date..

As our numbers have grown, so has the sophistication of our technology. Early humans made tools from materials readily to hand. Today's technology involves much more complex processes. As a result, we produce ever-increasing amounts of materials that pollute our air, soil and waterways.

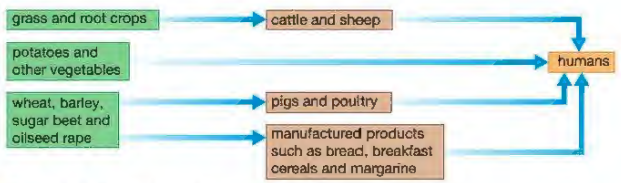
Early humans influenced their environment, but the enormous size of the population today and the extent of our industries mean that we affect the environment much more significantly. We make increasing demands on the environment for:

- food to sustain an ever-increasing population
- materials to build homes, schools and industries

- fuel to heat homes and power vehicles
- space in which to build homes, schools and factories, as well as for leisure facilities
- space in which to dump our waste materials.

MODERN AGRICULTURE – PRODUCING THE FOOD WE NEED

A modern farm is a sort of managed ecosystem. Many of the interactions are the same as in natural ecosystems. Crop plants depend on light and mineral ions from the soil as well as other factors in the environment. Stock animals (sheep, cattle and pigs) depend on crop plants for food (see Figure 15.3).



▲ Figure 15.3 A food web on a farm

Farmers must make a profit from their farms. To do this, they try to control the environment in such a way as to maximise the yield from crop plants and livestock.

IMPROVING YIELDS FROM CROP PLANTS

DID YOU KNOW?

Soil pH can vary between about pH 3 to 8. Soil pH is tested using indicator kits or pH meters. Some soils are too acidic – they can be made more alkaline by adding lime.



▲ Figure 15.4 (a) A glasshouse maintains a favourable environment for plants. Crops grown by hydroponics in a glasshouse.

Table 15.1 summaries some factors that can be controlled by the farmer in order to maximise crop yield.

Table 15.1 Some ways the yield from crops can be improved.

Factor controlled	How it is controlled	Reason for controlling the factor
soil ions (e.g. nitrates)	adding fertilisers to the soil or growing in a hydroponic culture (Figure 15.4a)	extra mineral ions can be taken up and used to make proteins and other compounds for growth
soil structure	ploughing fields to break up compacted soil; adding manure to improve drainage and aeration of heavy, clay soils	good aeration and drainage allow better uptake of mineral ions (by active transport) and water
soil pH	adding lime (calcium salts) to acidic soils; few soils are too alkaline to need treatment	soil pH can affect crop growth as an unsuitable pH reduces uptake of mineral ions
carbon dioxide, light and heat	these cannot be controlled for field crops but in a glasshouse or polytunnel all can be altered to maximise yield of crops (Figure 15.4b); burning fuels produces heat and carbon dioxide	all may limit the rate of photosynthesis and the production of the organic substances needed for growth

Glasshouses (otherwise known as 'greenhouses') and polytunnels can provide very controlled conditions for plants to grow (Figure 15.4). There are several reasons for this.

- The transparent walls of the glasshouse allow enough natural light for photosynthesis during the summer months, while additional lighting gives a 'longer day' during the winter.



▲ Figure 15.4 (b) Many crops are grown in large tunnels made of transparent polythene, called polytunnels.

- The 'greenhouse effect' doesn't just happen to the Earth, but also in greenhouses! Short wavelength infrared radiation entering the glasshouse is absorbed and re-radiated as longer wavelength infrared radiation. This radiation cannot escape through the glass, so the glasshouse heats up (see the section on the Earth's greenhouse effect later in this chapter). The glasshouse also reduces convection currents that would cause cooling.
- The glasshouse can be heated to raise the temperature if the outside temperature is too low.
- If heaters use fossil fuels such as gas, this produces carbon dioxide and water vapour. The carbon dioxide is a raw material of photosynthesis. The water vapour maintains a moist atmosphere and reduces water loss by transpiration.
- If the plants are grown in a hydroponic culture this provides *exactly* the right balance of mineral ions for the particular crop.

KEY POINT

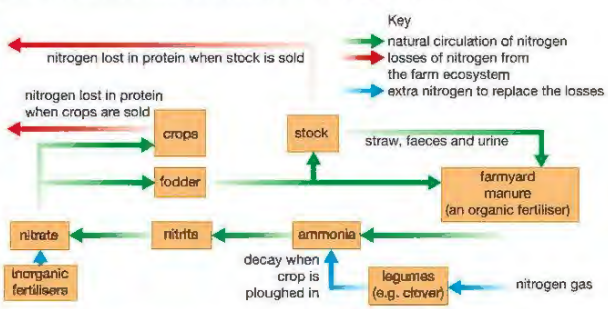
By heating glasshouses to the optimum temperature for photosynthesis, a farmer can maximise his yield. Heating above this temperature is a waste of money as there is no further increase in yield.

CYCLING NUTRIENTS ON A FARM

In Chapter 14 you saw how the elements nitrogen and carbon are cycled in nature. On a farm, the situation is quite different, particularly with regard to the circulation of nitrogen.

Nitrates from the soil supply nitrogen that is needed to make proteins in plants. Some of these plants are crops that will be sold; others are used as food for the stock animals (fodder). When the crops are sold, the nitrogen in the proteins goes with them and is lost from the farm ecosystem. Similarly, when livestock is sold, the nitrogen in their proteins (gained from the fodder) goes with them and is lost from the farm ecosystem. To replace the lost nitrogen, a farmer usually adds some kind of fertiliser. The amount of fertiliser added must be carefully monitored to ensure the maximum growth and yield of the crop – using excess fertiliser wastes money.

Figure 15.5 summarises the circulation of nitrogen on a farm.



▲ Figure 15.5 The nitrogen cycle on a farm. The effects of denitrification and lightning (see Chapter 14) have been omitted.

FERTILISERS INCREASE CROP YIELD

There are two main types of fertilisers – *organic* and *inorganic*. Many organic fertilisers (such as farmyard manure) are made from the faeces of farm animals mixed with straw. Inorganic fertilisers are simply inorganic compounds such as potassium nitrate or ammonium nitrate, carefully formulated to provide a

specific amount of nitrate (or some other ion) when applied according to the manufacturer's instructions.

Adding farmyard manure returns some nitrogen to the soil. But, as farmyard manure is made from livestock faeces and indigestible fodder, it can only replace a portion of the lost nitrogen. Most farmers apply inorganic fertilisers to replace the nitrates and other mineral ions lost. Whilst this can replace *all* the lost ions, it can also lead to pollution problems, which we will discuss later. Inorganic fertilisers do not improve soil structure in the way that organic fertilisers can, because they do not contain any decaying matter that is an essential part of the soil.

Another way to replace lost nitrates is to grow a legume crop (e.g. clover) in a field one year in four. Legumes have nitrogen-fixing bacteria in nodules on their roots (see Chapter 14). These bacteria convert nitrogen gas in the soil air to ammonium ions. Some of this is passed to the plants, which use it to make proteins. At the end of the season, the crop is ploughed back into the soil, and decomposers convert the nitrogen in the proteins to ammonia. This is then oxidised to nitrate by nitrifying bacteria and made available for next year's crop.

PEST CONTROL

Pests are organisms that reduce the yield of crop plants or stock animals. The 'yield' of a crop is the amount produced for sale. A pest can harm this in two ways:

- lowering the amount by reducing growth, e.g. by damaging leaves and reducing photosynthesis
- affecting the appearance or quality of a crop, making it unsuitable for sale (Figure 15.6).

DID YOU KNOW?
In Britain, about 30% of the potential maize crop is lost to weeds, insects and fungal diseases (Figure 15.6).



▲ Figure 15.6 Damage to a maize (corn) cob caused by the corn earworm caterpillar.

DID YOU KNOW?
A weed is a plant that is growing where it is not wanted. Weeds can be controlled mechanically or chemically. Mechanical control involves physically removing the weeds. Chemical control uses herbicides.

Any type of organism – plants, animals, bacteria, fungi or protoctists, as well as viruses – can be a pest. Pests can be controlled in a number of ways. Chemicals called **pesticides** can be used to kill them, or their numbers can be reduced by using **biological control** methods.

Pesticides are named according to the type of organism they kill:

- herbicides kill plant pests (they are weedkillers)
- insecticides kill insects
- fungicides kill fungi
- molluscicides kill snails and slugs.

A farmer uses pesticides to kill particular pests and improve the yield from the crops or livestock. Pests are only a problem when they are present in big enough numbers to cause economic damage – a few whiteflies in a tomato crop are not a problem; the real damage arises when there are millions of them. Whether or not a farmer uses pesticides is largely a decision based on cost. The increase in income due to higher yields must be set against the cost of the pesticides.

PROBLEMS WITH PESTICIDES

DID YOU KNOW?
Cultivating large areas of land with a single crop (a *monoculture*) encourages pests. Monocultures make harvesting the crop easier. But if a pest arrives, it can easily spread through the crop. During the winter the pest can lie dormant in the soil ready to attack next year's crop.
Crop rotation breaks the pest cycle. This is where a different crop is grown each year (Figure 15.7). When over-wintering pests emerge, their preferred crop is no longer there.



▲ Figure 15.7 A three-year crop rotation.

DID YOU KNOW?
DDT is a historical case, but there are problems with many modern pesticides. One recent example involves a class of insecticides called *neonicotinoids* (chemicals similar to nicotine). They have been linked to several serious ecological problems, including honey bee colony collapse disorder (CCD) and loss of birds due to a reduction in insect populations.

One problem with using pesticides is that a pest may develop *resistance* to the chemical. This happens through natural selection (see Chapter 19). It makes the existing pesticide useless, so that another must be found. Other problems are to do with the fact that pesticides can cause environmental damage. There are several reasons for this:

- they may be slow to decompose – they are *persistent* in the environment
- they build up in the tissues of organisms – *bioaccumulation*
- they build up and become more concentrated along food chains – *biomagnification*
- they kill other insects that are harmless, as well as helpful species, such as bees.

An ideal pesticide should:

- control the pest effectively
- be biodegradable, so that no toxic products are left in the soil or on crops
- be specific, so that only the pest is killed
- not accumulate in organisms
- be safe to transport, store and apply
- be easy to apply.

We can explain these problems by looking at one well-documented example – the insecticide DDT (dichlorodiphenyltrichloroethane).

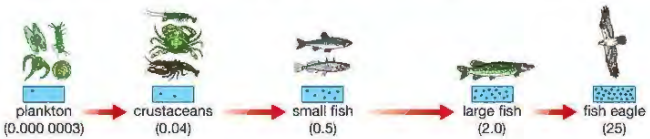
DDT was invented in 1874, and first discovered to be an insecticide in 1939. In World War II it was used with great success to kill malaria carrying mosquitoes and the lice that carried typhus. Its use increased up until the 1960s, when we began to understand its harmful effects. It was banned from general use in the USA in 1972, and worldwide in 2004. Limited use of DDT is still allowed for control of insects that transmit disease, although this is still regarded as controversial.

DDT is a very effective insecticide, so why has it been banned?

- DDT is very persistent, remaining active in the environment for many years. If DDT is sprayed onto a field, around half will still be there ten years later. To make things worse, the missing half won't have degraded to harmless products – some will have broken down to form a similar compound called DDE, which is also a potent insecticide, and some will have spread to other habitats. DDT is carried all around the world by wind, and has been identified in polar ice caps and deserts, thousands of kilometres away from where it was applied.
- By the 1950s, many types of insect began to appear that were resistant to DDT. These insects had developed a genetic mutation that prevented them from being killed by the insecticide. While DDT continued to be used, the resistant insects had an advantage over the non-resistant ones. Their numbers increased with every generation since they were able to survive exposure to the pesticide. They reproduced, passing on their resistance genes to their offspring. This is an example of **natural selection** (see

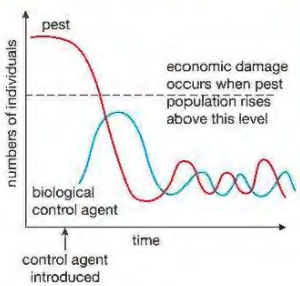
Chapter 19). There are now hundreds of examples of pest species that have developed resistance to different insecticides.

- DDT doesn't just kill pests. It will kill any type of insect, including harmless ones such as butterflies and useful species such as bees, as well as natural predators such as wasps, which might themselves kill the pest insects. Insecticides damage ecosystems by disrupting food chains.
- DDT is very soluble in fats. When a herbivore feeds on plants that are contaminated with DDT, the insecticide is not broken down or excreted. Instead it becomes concentrated in the fatty tissues of the animal. This is called **bioaccumulation**. When a carnivore eats the herbivore this process is repeated, so that the insecticide builds up in concentration along the food chain. This is known as **biomagnification** (Figure 15.8). The levels at the top of the food chain may be toxic, leading to the death of the top carnivores in the chain. This can disrupt the food web of an ecosystem.



▲ Figure 15.8 Biomagnification of DDT in a food chain

BIOLOGICAL CONTROL



▲ Figure 15.9 Biological control.

Another option that may be used to control pests is biological control. Instead of using a toxic chemical, biological control uses another organism to reduce the numbers of a pest. We have already mentioned whiteflies as pests of tomatoes. One way of controlling them in large glasshouses is to introduce a parasite that will kill the whiteflies. A tiny wasp called *Encarsia* parasitises and kills their larvae, reducing the numbers of adult whitefly.

A problem with biological control is that it never fully gets rid of a pest. If the control organism killed *all* the pests, then it too would die out, as it would have no food supply. Biological control aims to reduce pest numbers to a level where they no longer cause significant economic damage (Figure 15.9).

There are several methods of biological control. Some examples include:

- introducing a natural predator – ladybirds can be used to control the populations of aphids in orange groves
- introducing a herbivore – a moth was introduced from South America to control the prickly pear cactus that was becoming a serious weed in grazing land in Australia
- introducing a parasite – the wasp *Encarsia* is used to control whitefly populations in glasshouse tomato crops
- introducing a pathogenic (disease-causing) microorganism – the myxomatosis virus was deliberately released in Australia to control the rabbit population
- introducing sterile males – these mate with the females but no offspring are produced from these matings, so numbers fall
- using pheromones – these are natural chemicals produced by insects to attract a mate. They are used to attract pests (either males or females) to traps. The pests are then destroyed, reducing the reproductive potential of the population. Male-attracting pheromones are used to control aphids (greenfly) in plum crops.

BIOLOGY ONLY

FISH FARMING

Fish is a good source of high quality protein. Over the last 60 years, the world's population has demanded an increasing amount of fish to eat. The tonnage of fish caught by ships has increased steadily, while the Earth's stock of many fish species has decreased dramatically.

Increasingly, fish farming (aquaculture) is meeting the need for fish as a food supply. Farming of fish and shellfish is the fastest-growing area of animal food production. In 1970 only 5% of the world's seafood was produced by aquaculture, in 2016 it was 50%. The most commonly farmed fish are various types of carp, catfish, tilapia, trout, salmon, cod, bream and sea bass, as well as various types of crustaceans, such as lobsters and prawns. They are not all used for human food: about one-quarter of farmed fish is used to make animal feed.



▲ Figure 15.10 Feeding the fish at a fish farm

The fish are kept in densely stocked tanks or enclosures in rivers or lakes, or in sea cages (Figure 15.10). Fish farming has a number of advantages. The water quality can be monitored; for example the temperature, oxygen levels, water clarity and amount of chlorophyll in the water are measured. Large concentrations of chlorophyll give a warning of 'algal blooms' (see later in this chapter), which can be toxic to fish. Some conditions can be modified, e.g. air can be pumped into the enclosures to increase the amount oxygen dissolved in the water. The water is pumped through filtration units to remove the waste products of the fish.

The diet of the fish is also carefully controlled in both its quality and in the frequency of feeding. Enclosing the fish protects them against predators, and pesticides are used to kill parasites. Small fish may be eaten by larger members of their own species, so fish are regularly sorted by size and placed into different cages or tanks.

Selective breeding programmes (see Chapter 20) can be used to improve the quality of the fish. For example, they are bred to produce faster growth and to be more 'placid' (less aggressive) than wild fish.

However, fish farming has been heavily criticised by environmentalists. In any intensive production system, the potential for the spread of disease is greater than normal because the animals are so close together. Antibiotics are often used to treat disease. This is a cause for concern because the antibiotics may not have degraded by the time the fish are eaten by humans, adding to the problem of antibiotic resistance in bacteria (see Chapter 19). Fish farms also cause a pollution problem, producing organic material from the animals' faeces and from food pellets, which can contaminate the waters outside the fish farm and cause eutrophication of the water (see page 217). The pesticides used to kill fish parasites may be highly toxic to other non-harmful species of invertebrates.

In fact, there is strong evidence that fish farming has a negative effect on 'wild' fish stocks. Carnivorous species like salmon and sea bass are fed with pellets made from other fish. They need to eat several kilograms of wild fish to produce 1 kilogram of farmed fish! The wild fish used for fishmeal are less marketable species, such as herring and sardines.

END OF BIOLOGY ONLY

AIR POLLUTION

One definition of **pollution** is:

> 'Pollution is the contamination of the environment by harmful substances that are produced by the activities of humans'.

Human activities pollute the air with many gases. Some major examples that we will look at in this chapter are carbon dioxide, methane, carbon monoxide and sulfur dioxide.

CARBON DIOXIDE AND GLOBAL WARMING

KEY POINT

In any one year, there is a peak and a trough in the levels of carbon dioxide in the atmosphere

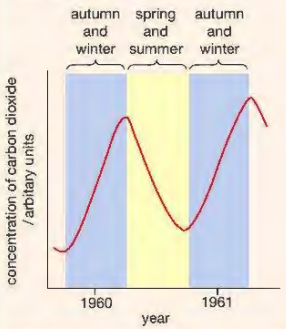
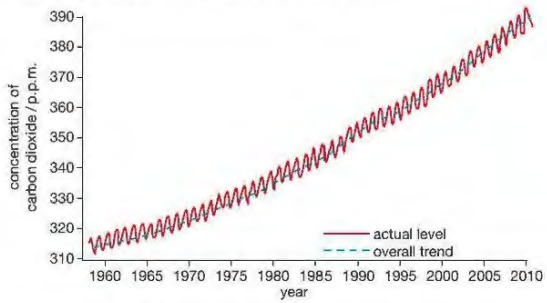


Figure 15.12 Seasonal fluctuations in carbon dioxide levels.

In the autumn and winter, trees lose their leaves. Without leaves they cannot photosynthesise and so do not absorb carbon dioxide. They still respire, which produces carbon dioxide, so in the winter months, they give out carbon dioxide and the level in the atmosphere rises. In the spring and summer, with new leaves and brighter sunlight, the trees photosynthesise faster than they respire. As a result, they absorb more carbon dioxide from the atmosphere than they produce, so the level decreases. However, because there are fewer trees overall, it doesn't quite return to the low level of the previous summer.

The levels of carbon dioxide have been rising for several hundred years. Over the last 100 years alone, the level of carbon dioxide in the atmosphere has increased by nearly 30%. This rise is mainly due to the increased burning of fossil fuels, such as coal, oil and natural gas, as well as petrol and diesel in vehicle engines. It has been made worse by cutting down large areas of tropical rainforest (see later in this chapter). These extensive forests have been called 'the lungs of the Earth' because they absorb such vast quantities of carbon dioxide and produce equally large amounts of oxygen. Extensive deforestation (see below) means that less carbon dioxide is being absorbed. Figure 15.11 shows changes in the level of carbon dioxide in the atmosphere (in parts per million) from 1960 to 2010.



▲ Figure 15.11 Changes in the level of carbon dioxide in the atmosphere.

The increased levels of carbon dioxide and other gases contribute to global warming, or the enhanced greenhouse effect. It is important to understand that the 'normal' greenhouse effect occurs naturally – without it, more heat would be lost into space and the surface temperature of the Earth would be about 30 °C lower than it is today, and life as we know it would be impossible.

Carbon dioxide is one of the 'greenhouse gases' that are present in the Earth's upper atmosphere. Other greenhouse gases include water vapour (H_2O), methane (CH_4), nitrous oxide (N_2O) and chlorofluorocarbons (CFCs). Most greenhouse gases occur naturally, while some (like CFCs) are only produced by human activities.

The 'normal' greenhouse effect is shown in Figure 15.13..

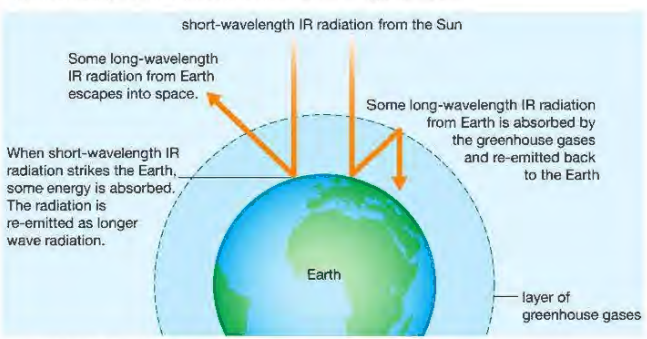


Figure 15.13 The Greenhouse effect

DID YOU KNOW?

Chlorofluorocarbons or CFCs are complex organic molecules containing carbon, chlorine and fluorine. They were once widely used in fridges, spray cans and as solvents, as well as in making materials such as foam packaging. Their use was banned or phased out in many countries in 1987, when they were found to be damaging the ozone layer in the upper atmosphere. (This layer protects living organisms by absorbing harmful ultraviolet radiation from the Sun.)

DID YOU KNOW?

Infrared radiation is radiated heat. The heat you can feel from an electric fire is IR radiation.

Short-wavelength infrared (IR) radiation from the Sun reaches the Earth. Some is absorbed by the Earth's surface and emitted again as longer-wavelength IR radiation. The greenhouse gases absorb and then re-emit some of this long-wavelength IR radiation, which would otherwise escape into space. This then heats up the surface of the Earth.

The problem is that human activities are polluting the atmosphere with *extra* greenhouse gases such as carbon dioxide. This is thought to be causing a rise in the Earth's surface temperature – the enhanced greenhouse effect, or global warming.

A rise in the Earth's temperature of only a few degrees would have many effects.

- Polar ice caps would melt and sea levels would rise.
- A change in the major ocean currents would result in warm water flowing into previously cooler areas.
- A change in global rainfall patterns could result. With a rise in temperature, there will be more evaporation from the surface of the sea, leading to more water vapour in the atmosphere and more rainfall in some areas. Other areas could experience a decrease in rainfall. Long-term climate change could occur.
- It could change the nature of many ecosystems. If species could not migrate quickly enough to a new, appropriate habitat, or adapt quickly enough to the changed conditions in their original habitat, they could become extinct.
- Changes in farming practices would be necessary as some pests became more abundant. Higher temperatures might allow some pests to complete their life cycles more quickly.

METHANE

Methane (CH_4) is an organic gas. It is produced when microorganisms ferment larger organic molecules to release energy. The most significant sources of methane are:

- decomposition of waste buried in the ground ('landfill sites'), by microorganisms
- fermentation by microorganisms in the rumen (stomach) of cattle and other ruminants
- fermentation by bacteria in rice fields.

DID YOU KNOW?

Cattle can produce up to 40 dm^3 of methane per animal per hour.

Methane is a greenhouse gas, with effects similar to carbon dioxide. Although there is less methane in the atmosphere than carbon dioxide, each molecule has a much bigger greenhouse effect.

CARBON MONOXIDE

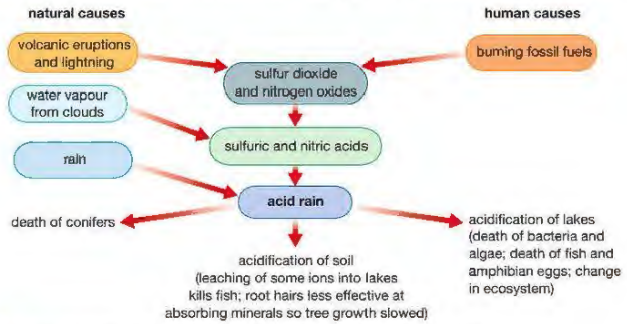
When substances containing carbon are burned in a limited supply of oxygen, carbon monoxide (CO) is formed. This happens when petrol and diesel are burned in vehicle engines. Exhaust gases contain significant amounts of carbon monoxide. It is a dangerous pollutant as it is colourless, odourless and tasteless and can be fatal. Haemoglobin binds more strongly with carbon monoxide than with oxygen. If a person inhales carbon monoxide for a period of time, more and more haemoglobin becomes bound to carbon monoxide and so cannot bind with oxygen. The person may lose consciousness and eventually die, as a result of a lack of oxygen reaching the cells, so that organs such as the heart and brain stop working.

KEY POINT

In many countries there are now strict laws controlling the permitted levels of carbon monoxide in the exhaust gases produced by newly designed engines.

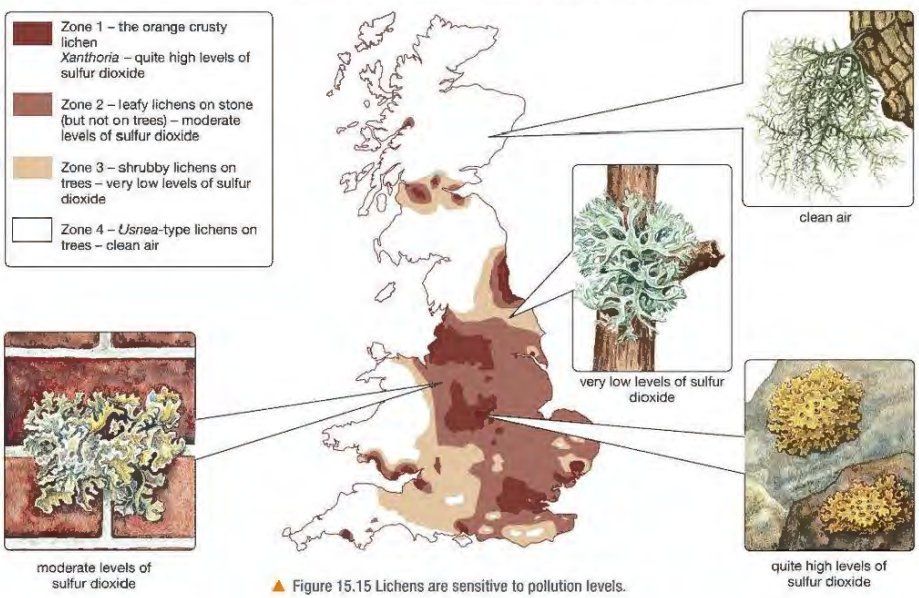
SULFUR DIOXIDE

Sulfur dioxide (SO_2) is an important air pollutant. It is formed when fossil fuels are burned, and combines with water droplets in the air. It can be carried hundreds of miles in the atmosphere before falling as **acid rain**. Rain normally has a pH of about 5.5 – it is slightly acidic due to the carbon dioxide dissolved in it. Both sulfur dioxide and nitrogen oxides dissolve in rainwater to form a mixture of acids, including sulfuric acid and nitric acid. As a result, the rainwater is more acidic with a much lower pH than normal rain (Figure 15.14).



▲ Figure 15.14 The formation of acid rain and its effects on living organisms.

Lichens are small moss-like organisms. Some lichens are more tolerant of sulfur dioxide than others. In some countries, patterns of lichen growth can be used to monitor the level of pollution by sulfur dioxide. The different lichens are called *indicator species* as they 'indicate' different levels of sulfur dioxide pollution.



▲ Figure 15.15 Lichens are sensitive to pollution levels.

The map in Figure 15.15 shows zones in Britain where different types of lichen are found.

BIOLOGY ONLY

DEFORESTATION

The last great natural forests of the world are the **tropical rainforests**. These form a belt around areas of the Earth near the equator, in South America, central Africa and Indonesia. They have a very high biodiversity. The rainforests are rapidly being destroyed by humans, in a process called deforestation. Deforestation is a consequence of the enormous growth of the human population. Every year, tens of thousands of hectares of rainforest are cut down to provide wood (timber) for building or other purposes, or to clear the land for farming (Figure 15.16). Much of the clearing is done by 'slash and burn' methods, where trees are cut down and burned, This adds to the carbon dioxide in the atmosphere and contributes to global warming. It also removes the trees, which would otherwise be absorbing carbon dioxide for photosynthesis. Deforestation is adding to global warming and climate change.



▲ Figure 15.16 Rainforest destroyed by the 'slash and burn' method.



▲ Figure 15.17 This river in Madagascar is full of silt from erosion caused by deforestation.

Apart from adding to global warming, there is a wide range of other problems caused by deforestation – some of these are listed below.

- Destruction of habitats and reduced biodiversity. Rainforests are home to millions of species of plants, animals and other organisms. It has been estimated that 50–70% of species on Earth live in rainforests.

- Reduced soil quality. There are no trees and other plants to return minerals to the soil when they die, and no tree roots to hold the soil together. Crops planted in deforested areas rapidly use up minerals from the soil, and rain washes the minerals out (leaching).
- The soil is exposed due to lack of tree cover (canopy), and is blown or washed away (Figure 15.17). Soil may be washed into rivers, causing rising water levels and flooding of lowland areas.
- Deforestation may produce climate change. Trees are an important part of the Earth's water cycle, returning water vapour from the soil to the air by transpiration through their leaves. Cutting down the forests will upset the water cycle.
- In the past, rainforests have been a valuable source of many medicinal drugs, as well as species of plant that have been cultivated as crops. There are probably many undiscovered drugs and crop plants that will be lost with the deforestation.

You can see that there are many reasons why conservation of the remaining rainforests is urgently needed. Sometimes controlled replanting schemes (reforestation or re-afforestation) are carried out, allowing for *sustainable* timber production. 'Sustainable' production means replacing the trees that are removed and ensuring that there is no ecological damage to the environment.

Tackling the problem of clearance of trees for farming is a more complex issue. Large-scale cattle farming on deforested land is carried out mainly to supply meat for the burger industry, or palm oil for cosmetics, and you may feel these reasons are not ethical ones. The small-scale farming by farmers around the edges of rainforests is more understandable. These farmers are poor, and their livelihood and that of their family depends on this way of life. The only alternative would be to resettle the farmers and give them financial help to establish farms in other areas.

DID YOU KNOW?

Cutting down and burning rainforests is adding to carbon dioxide in the atmosphere. But does it affect oxygen? Books often talk about the rainforests as being the 'lungs of the world' because of the large amount of oxygen they produce. But scientists think that overall they don't have much effect on the world's oxygen levels. In a rainforest, the oxygen produced by the living plants is roughly balanced by the oxygen consumed by decomposers feeding on dead plant material.

END OF BIOLOGY ONLY

WATER POLLUTION

Two major pollutants of freshwater are sewage and minerals from fertiliser.

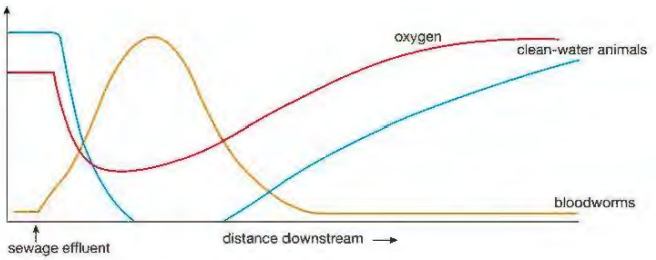
POLLUTION OF WATER BY SEWAGE

Sewage is wet waste from houses, factories and farms. In developed countries where large-scale sewage treatment takes place, industrial and agricultural sewage is usually dealt with separately from household sewage. Household sewage consists of wastewater from kitchens and bathrooms and contains human urine and faeces, as well as dissolved organic and inorganic chemicals such as soaps and detergents. It is carried away in pipes called sewers, to be treated before it enters waterways such as rivers or the sea.

If sewage is discharged untreated into waterways, it produces two major problems:

- Aerobic bacteria in the water polluted by the sewage use up the dissolved oxygen in the water as they break down the organic materials. This reduction in the level of oxygen kills larger animals such as freshwater insects and fish.
- Untreated sewage contains pathogenic bacteria, which are a danger to human health (see Chapter 13).

Where untreated or 'raw' sewage enters a river, the level of oxygen in the water becomes very low as the aerobic bacteria and other microorganisms from the sewage decompose the organic matter. Only species that are adapted to live in low-oxygen conditions, such as anaerobic bacteria, can survive. As the water moves away from the outlet, it becomes oxygenated again as it mixes with clean water and absorbs oxygen from the air. The increase in dissolved oxygen levels allows more clean-water species to survive. Figure 15.18 shows the changes in oxygen content, numbers of clean-water animals, and numbers of polluted-water animals downstream from a sewage outlet.



▲ Figure 15.18 Changes in oxygen levels and types of organism living downstream from a sewage outlet into a river.

The aim of sewage treatment is to remove solid and suspended organic matter and pathogenic microorganisms, so that cleaner waste can be discharged into waterways.

As with air pollution by sulfur dioxide, the level of pollution by organic material can be monitored by the presence or absence of indicator species. Figure 15.19 shows some of these.



▲ Figure 15.19 Some freshwater animals will only live in very clean water, while others can survive in very polluted areas.

EUTROPHICATION

Eutrophication comes from a Greek word, meaning 'well-fed'. It refers to a situation where large amounts of nutrients enter a body of water, such as a river, lake or even the sea. The nutrients in question are inorganic mineral ions, usually nitrates or phosphates. Pollution by these minerals can have very harmful effects on an aquatic ecosystem.

There are two main sources of excess minerals:

- from untreated or treated sewage
- from artificial nitrate or phosphate fertilisers.

Eutrophication is often caused by the use of artificial fertiliser. Streams and rivers that run through agricultural land that have been treated with fertiliser can contain high concentrations of nitrate and phosphate. This is because

nitrate is very soluble in water, and is easily washed out of the soil by rain, a process known as **leaching**. This is less of a problem with phosphate fertiliser, but phosphate is also washed into waterways by surface run-off of water.

The excess mineral ions stimulate the growth of all plants in the river or lake, but this is usually seen first as a rapid growth of algae, called an **algal bloom**. The algae can increase in numbers so rapidly that they form a thick scum on the surface of the water (Figure 15.20).



Figure 15.20 An algal bloom caused by fertiliser

REMINDER

The sequence of events following eutrophication is:

increase in mineral ions
↓
algal bloom
↓
death of algae
↓
decomposition by aerobic bacteria
↓
bacteria use up oxygen
↓
fish and other animals die

The algae soon start to die, and are decomposed by aerobic bacteria in the water. Because the bacteria respire aerobically, they use up the oxygen in the water. In addition, the algae block the light from reaching other rooted plants, further decreasing the oxygen produced by photosynthesis. The low oxygen levels can result in fish and other aerobic animals dying. In severe cases, the water becomes *anoxic* (containing very little oxygen) and smelly from gases like hydrogen sulfide and methane from the bacteria. By this stage only anaerobic bacteria can survive.

Rapid eutrophication is less likely when farmers use organic fertiliser (manure). The organic nitrogen-containing compounds in manure are less soluble and so are leached less quickly from the soil.

CHAPTER QUESTIONS

More questions on human influences on the environment can be found at the end of Unit 4 on page 221.

SKILLS CRITICAL THINKING 5

1 Which of the following is a desirable property for a pesticide?

A It kills all insects

B It does not show bioaccumulation

C It is persistent in the environment

D It is not biodegradable

SKILLS CRITICAL THINKING 5

2 The following gases are found in atmospheric air:

1. nitrogen
2. methane
3. carbon dioxide
4. water vapour

Which of these is/are *not* a greenhouse gas?

A 1 only **B** 1 and 4

C 2 and 3 **D** 2 and 4

BIOLOGY ONLY

SKILLS CRITICAL THINKING 5

3 Which of the following is a consequence of deforestation?

A soil erosion

B increased water vapour in the air

C reduced CO_2 in the air

D increased biodiversity

END OF BIOLOGY ONLY

4 In a lake affected by eutrophication, which of the following causes the death of aquatic plants?

A Poisoning by excess minerals

B Lack of oxygen in the water

C Lack of light reaching the plants

D Lack of carbon dioxide in the water

SKILLS REASONING 6

5 Why are humans having much more of an impact on their environment now than they did 500 years ago?

SKILLS ANALYSIS 4

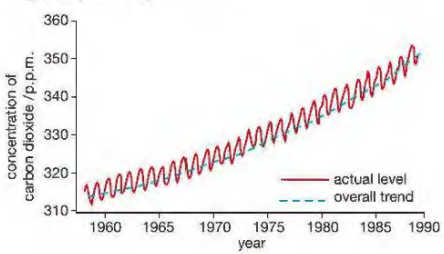
SKILLS REASONING 5

6 The graph shows the changing concentrations of carbon dioxide at Mauna Loa, Hawaii over a number of years.

a Describe the overall trend shown by the graph.

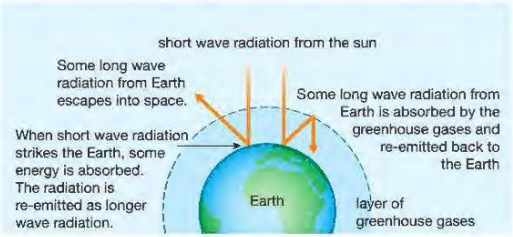
b Explain the trend described in a).

6 c In any one year, the level of atmospheric carbon dioxide shows a peak and a trough. Explain why.



SKILLS REASONING

7 The diagram shows how the greenhouse effect is thought to operate.

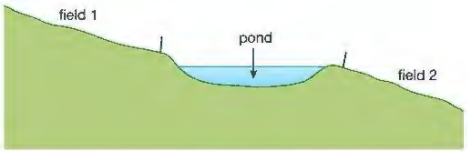


a Name two greenhouse gases.

b Explain one benefit to the Earth of the greenhouse effect.

c Suggest why global warming may lead to malaria becoming more common in Europe.

8 The diagram shows the profile of the ground on a farm either side of a pond.



The farmer applied nitrate fertiliser to the two fields in alternate years. When he applied the fertiliser to Field 1, the pond often developed an algal bloom. This did not happen when fertiliser was applied to Field 2.

a Explain why an algal bloom developed when he applied the fertiliser to Field 1.

b Explain why no algal bloom developed when he applies the fertiliser to Field 2.

c Suggest why the algal bloom is greater in hot weather.

9 Some untreated sewage was accidentally discharged into a small river. A short time afterwards, a number of dead fish are seen downstream of the point of discharge. Explain, as fully as you can, how the discharge might have led to the death of the fish.

10 Some farmers use pesticides and fertilisers to improve crop yields. Those practising 'organic' farming techniques do not use any artificial products.

a Describe how the use of pesticides and fertilisers can improve crop yields.

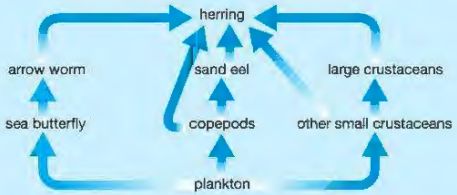
SKILLS CRITICAL THINKING

b Explain how organic farmers can maintain fertile soil and keep their crops free of pests.

UNIT QUESTIONS

SKILLS INTERPRETATION

1 The diagram shows a simplified food web of a fish, the herring.



a i Write out a food chain from the above food web containing four organisms. (1)

SKILLS ANALYSIS

ii From your food chain, name the primary consumer and secondary consumer. (2)

iii Name one organism in the web that is both a secondary consumer and a tertiary consumer. Explain your answer. (2)

b The amount of energy in each trophic level has been provided for the following food chain. The units are kJ per m^2 per year

plankton (8869) → copepod (892) → herring (91)

SKILLS INTERPRETATION

i Sketch a pyramid of energy for this food chain. (1)

SKILLS PROBLEM SOLVING

ii Calculate the percentage of energy entering the plankton that passes to the copepod. (2)

iii Calculate the percentage of energy entering the copepod that passes to the herring. (2)

iv Calculate the amount of energy that enters the food chain per year if the plankton use 0.1% of the available energy. (2)

SKILLS REASONING

v Explain two ways in which energy is lost in the transfer from the copepod to the herring. (2)

Total 14 marks